25900338 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACCFS200 - E-WASTE MANAGEMENT AND RECYCLING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Evaluate pure metals and metal oxides [E] / [CO4]
2. Illustrate the name of one biological method to enhance phytoremediation [An] / [CO2]
3. Differentiate electrotwinning and electrorefining [A] / [CO3]
4. Discuss about electrodeposition? [U] / [CO1]
5. Select the potential concern which is related to Phytovolatilization [E] / [CO4]
6. Classify types of contaminants into a table [U] / [CO1]
7. Identify which type of contaminants can phytoremediation address [U] / [CO1]
8. Identify the process used to selectively precipitate metals from solution in hydrometallurgy [A] / [CO3]
9. Write the process involves in conversion of solid into soluble ions? [A] / [CO3]

10. Illustrate the type of nanomaterials used to improve phytoremediation efficiency [A] / [CO3]

11. Identify an innovative way to reduce E-waste [U] / [CO1]

12. Predicate, how genetic engineering can improve phytoremediation efficiency [An] / [CO2]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Evaluate the main disadvantages and challenges of electrokinetic remediation. [E] / [CO4]

14. Write oxidation and reduction reactions with suitable chemical equations. [A] / [CO3]

15. Classify WEEE into five categories with examples [U] / [CO1]

16. Compare landfill and biological method [An] / [CO2]

17. Identify the major limitations of rhizofiltration in terms of type or location of contaminants [U] / [CO1]

18. Describe five metals in printed circuit boards and their importance [U] / [CO1]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Explain adsorption and desorption processes in detail. Describe their applications in e-waste management with examples. [U] / [CO1]

20. Illustrate the way in which phytodegradation can be improved [A] / [CO3]

21. Evaluate the practical applications and limitations of electrokinetic remediation in large-scale organic waste treatment. [E] / [CO4]

[10x2 = 20]